

Epistemology in a Nutshell

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Can we be sure about our Beliefs?

- Our **claims** are supported by **Argumentation**
- Argumentation:
 - Premises (facts and assumptions)
 - Reasoning

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But, **when can we be sure that our claim is true**, if we can be sure of anything at all?

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Epistemology is the branch of philosophy that examines the nature, origin, and limits of knowledge

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Three Philosophical questions:

- What is knowledge?
- How can be sure about what is true?
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Having the answer to these questions is fundamental to Science

Science

Science is about discovering **new true** knowledge and **making it available** to the world

Some Milestones and Keywords in the History of (Philosophy of) Science

Greek Scholars, 400BC

- Thinking about what is to think
- Became aware of our cognitive processes
- Look for **Natural explanation of phenomena** as opposed to:
 - predictions by correlation
 - divine explanations

Predictions by Correlation

September in Barcelona

September is a rainy month in Barcelona

Divine Explanation of Phenomena

Why it always rain in La Mercè's day (sep 24th)?

It is *Santa Eulàlia* crying, because she is the Saint patron of Barcelona and feels left behind by Barcelonians who prefer *La Mare de Deu de la Mercè*

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There may be a lot of useful (i.e, practical) wisdom based on divine explanations, but only when you understand the **natural explanation** you can really exploit nature.

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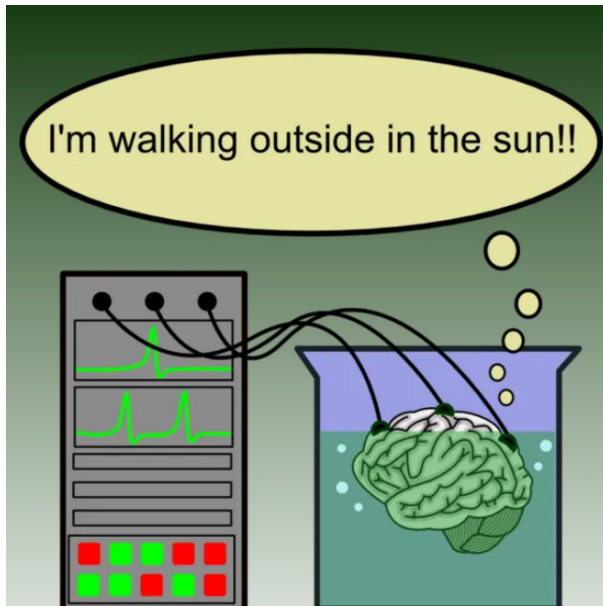
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Example: There are many cultural/religious cleansing, which have good consequences (hygiene is good), but only when you understand germs you fully understand hygiene.

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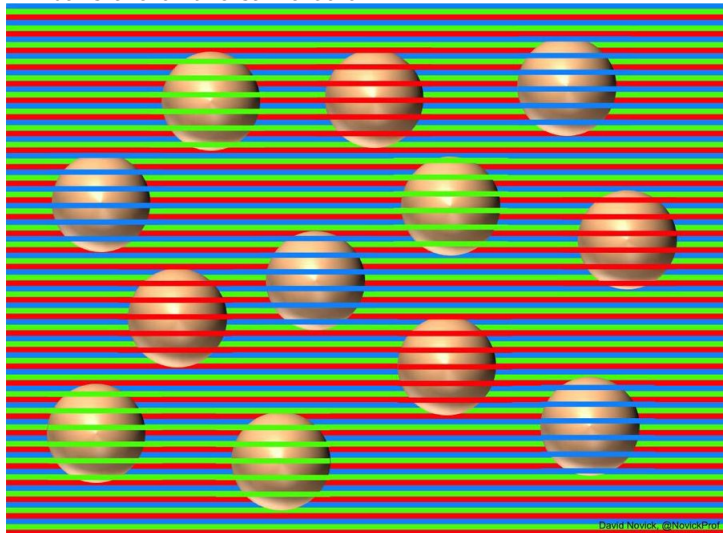
- **Greek Scholars** (400 BC).
 - **Plato**. Myth of the cave
 - We cannot trust observation (our perceptions are fictitious)
 - **Rationalism**: Knowledge can only be gained by **reasoning**

Can we trust what we Observe?



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All balls are of the same color



David Novick, @NovickProf

Some Milestones and Keywords in the History of (Philosophy of) Science

- **Greek Scholars** (400 BC).
 - **Aristotle.**
 - Premises are learned from observation
 - **Empiricism:** Knowledge has to be gained by **observation** (and reasoning)
 - First formalization of **reasoning** (Formal Logic, deductive reasoning)
 - (Implicit) **deductive** reasoning

Some Milestones and Keywords in the History of (Philosophy of) Science

Enlightenment (XVII)

Sapere Aude (dare to think by yourself)

- Mankind instead of God is at the **center** of the universe
- Have the courage to use your own understanding

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Galileo:

- Separate Science from Theology
 - If empirical evidence contradicts religion, religion cannot censor it

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 - If empirical evidence contradicts religion, religion cannot censor it
- **Systematic Observation** (systematically record data about the phenomenon under study before assuming any thing)
- (Implicitly) **Inductive** reasoning

The importance of systematic observation: Can we trust what we Remember?

[See details here](#)

Memories from 9/11 (largest scientific study on collective memory)

- How did you learned about the attack?. Accuracy 63% one year later, 57% three years later
- What were your emotions?. Accuracy 40% one year later
- Quote from the study's author: "You can't even convince people that their memories are wrong"
- Remember **confirmation bias**: we tend to ignore data that does not fit to our believes

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 - Philosophical crisis

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A case for Techno-optimists

Some people say that we shouldn't worry about climate change because technology will save us as it has always done in the past. Skepticism says that what happened in the past does not guarantee the future.

Some Milestones and Keywords in the History of (Philosophy of) Science

Modern Science (1900).

- **Logical Positivism** (Wien Circle).
 - Science= study of **meaningful** statements
 - **Meaningful**: it must be possible to determine the truth (**verification principle**)
 - Two types of meaningful statements:
 - **Analytic** (definitions+ verifiable by formal logic)
 - **Synthetic** (verifiable by observation)

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Karl Popper (Wien 1902)

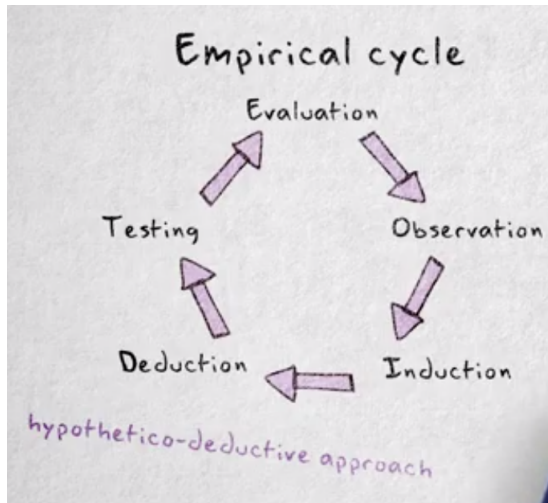
- **Logical Empiricism.** A statement is useful if it is **falsifiable**.
 - Replace verification principle by **confirmation**
 - Science is about statements that can be confirmed or falsified

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 - Science is about statements that can be confirmed or falsified
- **Constructive Empiricism.**
 - Science attempts to construct theories that explain empirical results
 - Scientific claims are taken as true only as far as the observations go along (i.e, claims can be at most **temporarily** true)
 - **Hypothetic-deductive Scientific Method**

(Hypothetic-Deductive) Scientific Method



- ① **Observe** and try to make sense out of what you observe
- ② **Inductive step**: from observation and thoughts come out with a hypothesis (let your intuition work)
- ③ **Deductive step/experimental design**: make a prediction that should happen if the hypothesis is true (in other words, a prediction that if does not happen would make your hypothesis highly unlikely)
- ④ **Testing step/experiment**: check if the prediction came true
- ⑤ **Evaluation**: if prediction/exp. outcome goes along hypothesis, then increase likelihood of hypothesis. Else revise or reject hypothesis

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p-value

If you make the experiment **statistically meaningful**, you end up with a *p*-value (typically, if $p < 0.05$ you can accept your hypothesis).

Am I my cat's favorite?

- **Observation:** My cat shows me so many signs of affection
- **Hypothesis** (Inductive step): My cat prefers me over my wife
- **Test design** (Deductive step):
 - find an observable (testable) variable that we believe is a proxy of cat's love (objective)
 - design an experiment (testable, replicable, can falsify the hypothesis,...)
- **Testing:** Conduct the experiment
- **Evaluate** the results and see how well they explain the hypothesis

Example: The Titius-Bode Law

J.E. Bode (XVIII German Astronomer)

- Claimed that the **distance of planets** w.r.t. sun follows a **geometric progression** of power 2 (and gave the detailed function)
- The Law, which fitted the known planets, was considered (moderately) interesting

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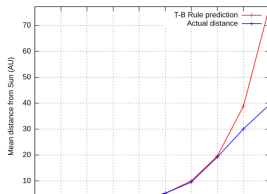
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- It was widely accepted at that point, until in 1846 Neptune was discovered in a location that does not conform to the law
- The discovery of Pluto in 1930 confounded the issue still further
- Now it is considered a spurious curiosity



Non Scientific support for a Hypothesis

- **Intuition/Believe** (we are too irrational)
- **Consensus** (we are too inclined to follow the crowd)
- **Authority** (we are too lazy to do the thinking on our own)
- **Casual observation** (we cannot trust our memory and we suffer from confirmation bias)
- **Weak argumentation** (recall lecture on Critical Thinking)

The previous methods are weak, subjective and very prone to errors.

The Scientific Method is socially overrated

- It is just a methodological **template**
- It assumes that we can trust our senses, we can make measurements and we can learn from sensory experience
- It is supported by many **success** stories (i.e, it does the job)
- It can make **mistakes** (and has done so many times)
- It is not true that researchers just neatly apply the method (researchers activities are much more complex and dynamic than that)
- There are many examples of breakthrough results produced by **chance** and/or **accident**.

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Scientific Attitude

- **Open** (willing publicise work)
- **Critical** (willing to give and get constructive feedback)
- **Transparent** (willing to allow replicability)

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Public research fits better than **Private** research

- The importance of the Scientific Method is overrated
- The importance of the Scientific Attitude is underrated

Back to the Scientific Method: Verification vs. Falsification

- **Verification:** This involves seeking evidence to confirm a theory. Popper argued this is often too easy and not a rigorous way to test a theory.
- **Falsification:** This involves seeking evidence that disproves a theory.

Popper

Our confidence in a theory isn't based on how many times it's been proven right, but on how many times it has been rigorously tested and failed to be proven wrong

- A better name for the verification step of the Sci.Method is **Failed Falsification** step.
- Doing this requires **critical thinking** and **intellectual integrity**

Scientific Method and Attitude

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- must always keep a certain level of skepticism
- That's what distinguishes Science from pseudo-science or negationism

Negationism and Pseudo-science

- Pseudo-scientists *see what they believe*, instead of *believing what they see*.
- A decision is made beforehand about what one wants to be true.
- Then high acceptance standards are required for the things that they do not want to believe and very low for the things that they want to believe

Nobel prize in Economics 2021

The prize was awarded to David Card, Joshua Angrist and Guido Imbens for real-world research in the 1990s that demonstrated, empirically, that the idea touted by conservative economists that higher minimum wages mean fewer jobs is not based on fact.

The highest award is given for demonstrating that the community was wrong.

Scientific Attitude: Failure Example

Michael Siegel, public-health researcher at Tufts

- Most of the research which has been funded by alcohol companies has reported that there is a benefit to drinking moderate amounts of alcohol.
- Most of the research that has not been funded by the industry has found that there actually is not a benefit, and that overall mortality is higher.

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- The same thing happened with cigarettes, and sugar
 - Hopefully, good science will eventually make his way on this issue as well

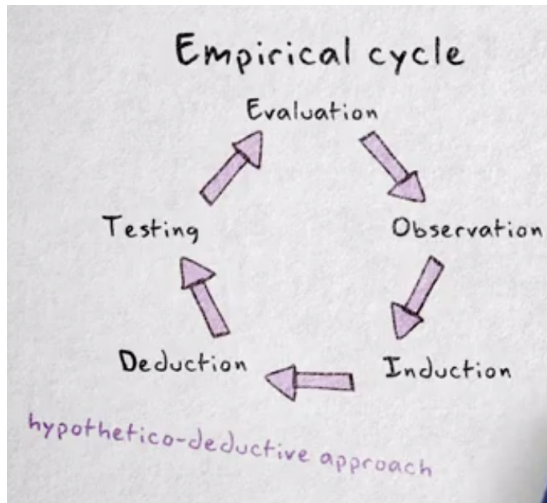
Scientific Attitude: Another Failure Example

Sample size in experiments

- When researchers develop a new treatment, they want the treatment to be effective
- Treatment effectiveness is shown experimentally (Scientific Method)
- Having volunteers for the experiment is expensive. Therefore, it is not unusual that the experiment is started with small samples, which are increased until results produce the right p -value
- Can you see why this procedure may be problematic? It gives advantage to proving a false positive.

Scientific Method, Belief Updating

We have seen that scientists need to revise their beliefs



An important aspect of the Method is how we update our beliefs

Bayes Theorem

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$$P(\mathcal{H}|\mathcal{D}) = \frac{P(\mathcal{D}|\mathcal{H}) \times P(\mathcal{H})}{P(\mathcal{D})}$$

Priors $P(\mathcal{H})$, **likelihood** $P(\mathcal{D}|\mathcal{H})$, **posterior** $P(\mathcal{H}|\mathcal{D})$

Example

The prevalence of colon cancer (i.e, probability of having cancer) is 0.01. To diagnose Cancer (C) we have a test (T) which gives false positives with probability 0.2 and false negatives with probability 0.1. We would like to know what is the probability of a patient having cancer if the test was positive ($P(c|t)$)

Bayes Rule: example medical diagnosis

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we normalize because the two numbers must add up to 1

$$P(c|t) = 0.009 / (0.009 + 0.198) = 0.043$$

$$P(\neg c|t) = 0.198 / (0.009 + 0.198) = 0.957$$

Bayes Theorem, alternative (prior and posterior odds)

$$\frac{P(\mathcal{H}|\mathcal{D})}{P(\neg\mathcal{H}|\mathcal{D})} = \frac{P(\mathcal{D}|\mathcal{H})}{P(\mathcal{D}|\neg\mathcal{H})} \times \frac{P(\mathcal{H})}{P(\neg\mathcal{H})}$$

In the previous example,

$$\frac{P(t|c)}{P(t|\neg c)} = \frac{0.9}{0.2} = 4.5$$

The ratio has increased 4.5 times

Bayesian Belief update: Example

In many contexts, we do not have well-founded numbers, but we have subjective beliefs and can still use the theorem to give us some intuitions.

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Neanderthals

It is generally believed that Neanderthals DID NOT have symbolic thinking (symbolism may be manifested as figurative art, ornamentation, and ritual burial). But recently personal ornaments have been found near a Neanderthal archaeological site

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Disclaimer: I am making up the numbers

- $\frac{P(\mathcal{D}|\mathcal{H})}{P(\mathcal{D}|\neg\mathcal{H})} = \frac{0.1}{0.9} = 0.1$
- Because of the finding, the prior odds should be divided by 10

Check-out these two interesting examples:

- Lottery Fraud in Philippines
- Election Fraud in Venezuela